

Arya Satam

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EXPERIENCE

Programmer/Analyst I | The Research Foundation of State University of New York

Oct. 2024 - Present

- Implemented RAG-based clinical protocol evaluation using vector retrieval over regulatory guidance and LLM-based synthesis.
- Generated structured clinical protocol evaluations using prompt-driven LLM reasoning with explicit guidance citations.
- Built a production-ready backend with pgvector-enabled embedding retrieval in PostgreSQL for efficient semantic search.
- Centralized scattered clinical trial records into a single interface, enabling researchers to visualize drug interactions seamlessly.
- Redesigned and migrated from Flask to PERN stack to improve scalability and enable prioritization of concurrent requests.
- Streamlined ETL workflow with Python, SQL, & UNIX command line, improving data quality & cutting processing time by 60%.
- Optimized PostgreSQL queries with indexing and caching, for faster access to patient records and therapeutic datasets.
- Collaborated with cross-functional teams to translate research workflows into decision-support dashboards enabling causal interpretation, model attribution, and incremental outcome analysis.

Data Scientist | Skill Vertex

Jun. 2022 - Aug. 2022

- Performed advanced analytics on 54K entity-level records using Pandas/NumPy to optimize objectives & quality assurance.
- Developed predictive ML models (Random Forest, Logistic Regression, Decision Trees) optimizing precision-recall tradeoffs.
- Improved model performance by 15% through feature attribution analysis and controlled model comparisons.
- Applied statistical and propensity-based techniques to reduce bias and improve attribution of model-driven outcomes.
- Deployed secure, containerized systems via Docker and AWS (Lambda, S3, RDS) with CI/CD automation enhancing decision-making efficiency.

Data Scientist | Skill Safari

Jun. 2021 - Aug. 2021

- Built XGBoost model (85% accuracy) for loan approvals, identifying credit score/income as key drivers, reducing default rates.
- Implemented Neural Network model to predict cardiovascular risk, elevating early detection & at-risk patient awareness by 20%.
- Forecasted weather with 95% accuracy using ARIMA and LSTM models, optimizing agricultural and logistics planning.

PROJECTS

Toxic Comment Detection: Python, ML, NLP, Apache Airflow, Docker, AWS [\[code\]](#)

- Developed an NLP pipeline for toxic comment detection on 100K+ records, applying text annotation & semantic analysis.
- Automated ETL workflows with Apache Airflow to AWS S3, then deployed a containerized Flask API via AWS ECR & API Gateway, delivering a real-time EC2-hosted interface with 96% accuracy using DistilBERT.

Advanced ATS Analyzer: OpenAI API, NLP, Python, Streamlit, Prompt Engineering [\[code\]](#)

- Formulated a dashboard style AI-powered ATS resume analyzer using OpenAI GPT models, Streamlit, PyPDF2, and dotenv, enabling automated resume evaluation against job descriptions by extracting key insights and assessing technical skill matches.
- Elevated resume screening efficiency by implementing structured JSON feedback with skill match percentages, missing keywords, and experience relevance, improving ATS compatibility for job applications.

Umpire Sign Recognition: Python, CNN, OpenCV, TensorFlow [\[code\]](#)

- Built a real-time gesture-recognition pipeline using CNNs with ROI-based hand segmentation. Designed background-subtraction preprocessing to isolate hand movement across lighting conditions. Collected & labeled a custom dataset with a data-capture notebook and trained a CNN model on multi-class gestures. Executed a live prediction loop achieving real-time inference speeds for continuous hand-sign decoding.

CERTIFICATION

AWS Certified Cloud Practitioner [\[link\]](#)

EDUCATION

University at Buffalo

Master of Science in Engineering Science - Data Science

Aug. 2023 – Dec. 2024

New York, US

University of Mumbai

Information Technology

Aug. 2019 – May. 2023

Maharashtra, IND

SKILLS

Programming & Query Languages : Python, R, Java, C, C++, Go, MATLAB, JavaScript, TypeScript, SQL, NoSQL (MongoDB), Hive

Markup / Scripting / Command Line : HTML, CSS, XML, UNIX, Bash, Shell Scripting

Tools / Frameworks : TensorFlow, Keras, PyTorch, Scikit-learn, Big Data technologies (Apache Spark, Apache Airflow, Apache Hadoop, Hive) Hugging Face Transformers, LangChain, OpenAI API, BI Tools (Tableau, Power BI, Excel), Git, Streamlit, MLflow

Cloud Computing : AWS (S3, Lambda, Glue, RDS, DynamoDB, SageMaker), Azure, GCP, Snowflake, Databricks, Salesforce

Data Science and AI : Data Cleaning (Pandas, NumPy, PySpark), Data Visualization (Matplotlib), Data Manipulation, Data Mining, Clustering, ML, Predictive Analytics, Deep Learning, Decision Trees, Statistics, GenAI, NLP, Time Series Forecasting, RAG, ETL